Collaborative Filtering: Personality Diagnosis

Axel Belloir, Cenzo Poirier De Carvalho, Théodore Labrande

Groupe Blocus

15/10/2026

Baseline approach

- Two baselines were implemented so far
- Classic Matrix Factorization with GD
- Matrix Factorization with mean and bias using GD

Baseline results

The naive MF gives an RMSE of around 1.25 (on test), whereas the mean centered MF gives an RMSE around 0.867 (on server), already out-performing BestOf2024-2.

Collaborative Filtering by Personality Diagnosis

- 2000s article from researchers at Microsoft and NEC
- Uses users' ratings as priors for their true ratings, allowing to then compare users distributions to infer unseen ratings.

$$\begin{aligned}
&\Pr(R_{aj} = x_j \mid R_{a1} = x_1, \ldots, R_{am} = x_m) \\
&= \sum_{i=1}^{n} \Pr(R_{aj} = x_j \mid \mathbf{R}_a^{\text{true}} = \mathbf{R}_i) \\
&\quad \cdot \Pr(\mathbf{R}_a^{\text{true}} = \mathbf{R}_i \mid R_{a1} = x_1, \ldots, R_{am} = x_m)
\end{aligned}$$

Figure 1: The probability that user a rates product j as x_j

Assumptions

- Preferences can be modeled in a Bayesian way
- User ratings contain gaussian-modeled noise with parameter σ
- Cold start can be dealt with uniform distribution on the priors, i.e. by picking the mode

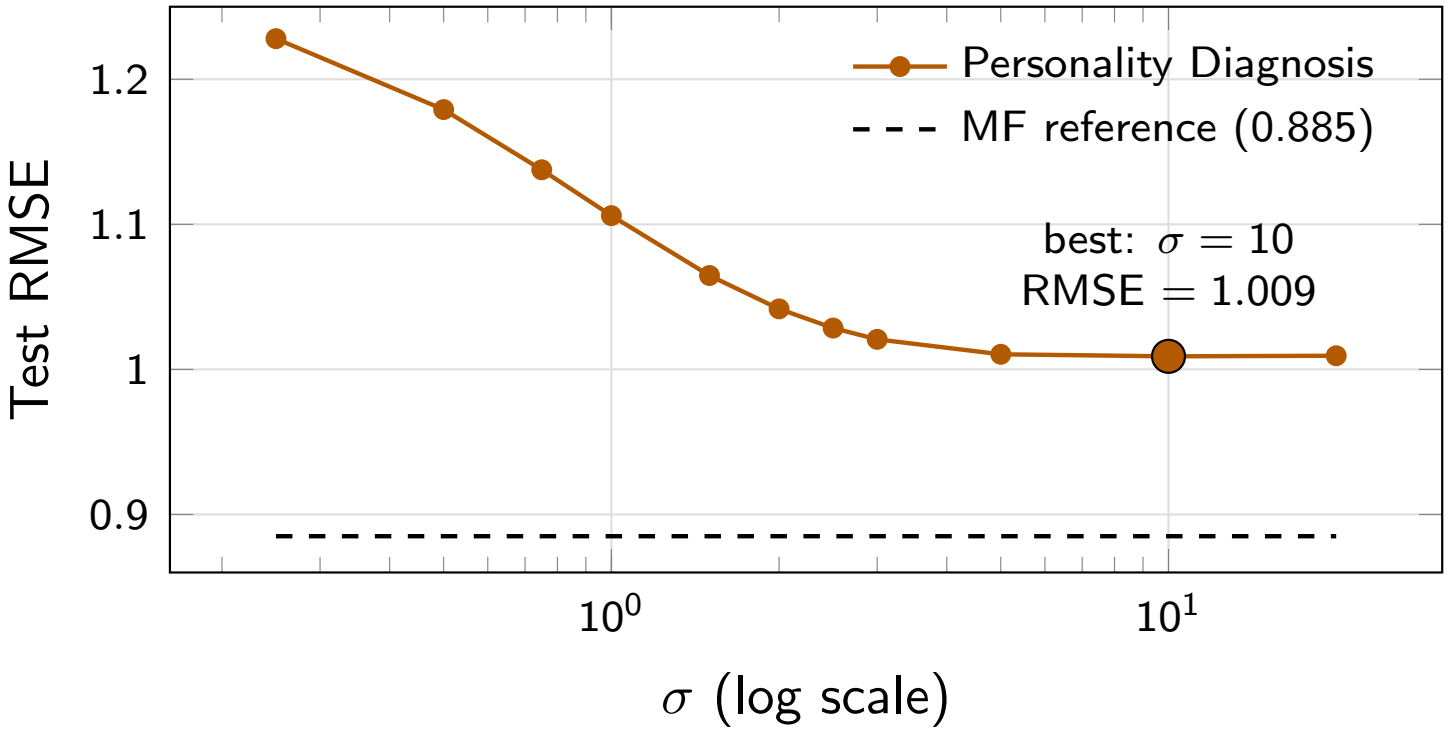
Study results

- Outperformed other methods at the time, especially with extreme ratings, all measured with MAD.
- Limited because products were removed when insufficient data

Our implementation vs. the paper

- **Prediction:** expected rating (mode also available) instead of the mode only, since we are scored with RMSE, not MAD.
- **Neighbors:** only users sharing at least 2 rated items with the active user.
- **No neighbor rated the item:** fallback to a bias baseline (global mean + user and item biases), not specified in the paper.
- **Predictions clipped** to the range of observed ratings.
- **Noise parameter σ:** swept from 0.25 to 20 (paper: hand-tuned to 2.5).
- **Protocol:** fixed train/test split scored with RMSE (paper: all-but-1 and given-N protocols, MAD).

Current results



Personality Diagnosis: test RMSE vs. σ

For $\sigma \geq 5$, the RMSE stays at ≈ 1.01.

For next week

Experimental evaluation

- Check train/test separation.
- Tune hyperparameters on validation, not test.
- Compare PD and MF using the same split and RMSE.

Why might PD underperform MF?

- Sparsity: few shared ratings between users.
- Sensitivity: PD depends on the Gaussian noise parameter σ.
- Latent factors: MF captures indirect user–item relationships.

Goal: validate the results and understand the performance gap.